The relationship between sleep and physical activity across the lifespan

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Abstract

Objectives: To examine the association between device-measured physical activity and sleep across the lifespan using individual participant data (IPD) meta-analysis. We furthed aimed to examine how the association differed by age.

Design: Pooled IPD analysis of accelerometer-based physical activity and sleep data from multiple observational studies.

Setting: We sourced data from 20 observational studies conducted multiple geographical regions and across the lifespan.

Participants: The dataset included 220,760 observations from 28,823 participants aged 2 to 94 years. Participants were selected from studies that used research-grade accelerometers with at least four valid days of wear time, including at least one weekend day. Exclusion criteria included insufficient wear time and missing key covariate data.

Main Outcome Measures: Sleep outcomes included total sleep time, sleep efficiency, sleep onset timing, and sleep regularity. Physical activity was assessed in terms of volume (average acceleration) and intensity (intensity gradient). Mixed-effects models were used to examine associations, adjusting for age, sex, body mass index, and socioeconomic status.

Results: Higher physical activity volume was associated with longer sleep duration, greater sleep efficiency, earlier sleep onset, and more regular sleep. The effect of physical activity on sleep regularity was particularly strong, with minimal variation across age groups. Conversely, individuals with more regular sleep patterns and earlier sleep onset exhibited higher levels of physical activity the following day, though these effects were modest.

Conclusions: Physical activity is positively associated with sleep regularity across the lifespan, supporting its role as a non-pharmacological intervention for maintaining consistent sleep patterns. Public health recommendations should emphasize the importance of regular

physical activity for sleep health. Future research should investigate mechanisms underlying these associations and assess targeted intervention strategies.

Trial Registration: The study protocol was pre-registered on the Open Science Framework (https://osf.io/ahd96).

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Introduction

Sleep is an important determinant of population health across the lifespan [1,2]. The consequences of poor sleep include increased all-cause mortality [3], cardiovascular disease [4], cancers [5], and poor mental health [6]. Beyond duration and quality, the regularity of sleep timing is now recognised as a distinct dimension of sleep health [7,8], and irregular sleep is consistently associated with poorer mental, cardiometabolic, and survival outcomes [9]. Poor sleep - characteristed by low sleep quality, duration, or the presence of sleep disorders - is a frequent health problem: 14-68% of children and adolescents [10] and 31% of adults [11] report lower sleep duration than recommended [12,13]. Improving sleep should be a priority for public health policy [14], and more safe and accessible strategies that promote sleep are needed [15]. In adults, including older adults, being physically active appears to improve both the quantity and quality of sleep [16–18], and exercise is an effective treatment for insomnia [19]. The World Health Organization recommends physical activity as a method to improve sleep in adults and older adults [20]. In children and adolescents, associations are less clear [15,21,22]: recent trials suggest exercise improves sleep quality and efficiency but not total sleep time [23], and guidelines do not currently support a role for physical activity to improve sleep in this age group.

Promoting physical activity may be an effective method to improve sleep across the lifespan, but the published literature has at least two limitations. First, most studies have relied on self-report measures of sleep and physical activity [15,21,22], which could confound associations and limit the precision of analyses designed to identify a dose-response relationship between physical activity and sleep. Second, no study has collected day-to-day data from a large and geographically diverse sample that included participants across the lifespan: daily associations have been synthesised separately in adults [24] and in children and youth [25], and recent consumer-wearable studies with millions of person-nights were

restricted to adults and did not examine sleep regularity [26,27]. Umbrella reviews [15] and meta-analyses [4,5,16,21,22,28–43] have provided some suggestive evidence about how physical activity may improve sleep across the lifespan. However, their reliance on aggregated study-level effects limits the quality and scope of data that can be used to analyse the influence of age [44]. Device-measured individual participant data analysis can overcome these limitations [45].

In this study we harmonised individual participant data from accelerometer observations obtained from studies around the world. The aim of this pooled study was to investigate the relationship between device-measured physical activity and sleep across the lifespan. We examined whether individuals who are more physically active have better sleep, whether age, sex, and socio-cultural factors moderate the association between physical activity and sleep, and whether there is a bidirectional association between physical activity and sleep.

Methods

We pre-registered all methods and analyses on the Open Science Framework (OSF) before data acquisition began (https://osf.io/ahd96).

Data sources and data acquisition

We included data from studies, with any design, that used research-grade accelerometers and a 24-hour wear time protocol over a minimum of 4 days, including 1 weekend day. Studies needed to have used accelerometers from which raw accelerations (m/s^2 or mg) could be obtained. Studies could use any wear location for the accelerometer (hip, wrist, or thigh) or accelerometer brand.

We identified potentially relevant studies by examining the reference lists of recent systematic reviews that explored the relationship between sleep and physical activity in children, adolescents, and adults [16,21,22]. We then used forward searching on potentially

eligible studies [46]. We additionally performed a literature search to identify accelerometry consortiums and individual studies, searched trial registries (ANZCTR, ICTRP, ISRCTN) and the Open Science Framework for unpublished results, and asked collaborators to help identify any additional studies.

We contacted chief investigators or consortium leads by email to request access to deidentified individual data. To make the data acquisition manageable, we only invited studies with sample sizes larger than 400 participants. If required by the contributing research group, the contributing investigators or their university signed a data transfer agreement. All studies included in the pooled data meta-analysis had to have been approved by their local or institutional ethical committees, and all participants had to have provided written informed consent. The lead university's Research Ethics Committee granted permission for each study's data to be obtained, and investigators transferred their deidentified data via a secure file-sharing system.

Data harmonisation and outcomes

All accelerometer-derived data were harmonised by reprocessing raw files using an open-source GGIR package in R to produce physical activity and sleep outcomes [47,48]. GGIR's sleep detection has been evaluated against polysomnography in both adults [48] and children [49,50]. By reprocessing all raw accelerometer data, we were able to generate outcomes that could be compared across studies, device wear locations, and populations [51]. Data were reprocessed by the core research team or the data contributors themselves using the GGIR package processing script provided by the core research team. Reprocessed individual study datasets were then pooled into a single dataset. A detailed description of the reprocessing method and decisions for specific accelerometer devices is available in an online supplementary file.

For sleep, GGIR produced the following metrics: total sleep time (minutes), sleep

efficiency after onset (percentage of time asleep within the sleep period), sleep regularity (regularity of sleep and awake patterns on a day-to-day basis calculated as the sleep regularity index), and timing (time of sleep and awakening within the 24-hour time cycle). For physical activity, we used volume and intensity metrics proposed by Rowlands et al [52]. These measures are 1) average acceleration (measure of physical activity volume) and 2) intensity gradient (measure of intensity distribution where the natural logs of time and intensity are used to produce a straight-line graph with the slope of the graph being the intensity gradient). We used this approach to model the entire continuum of physical activity intensity for our primary analyses [53] and avoid collapsing physical activity data into population-specific intensity categories, which occurs when intensity cut-points are applied to accelerometer data [54].

To translate physical activity data for interpretation and public health recommendations and for time-use based analyses we then determined; 1) the number of minutes spent in acceleration intensity bands in $50mg$ increments [52], 2) the number of minutes spent in light and moderate to vigorous physical activity using age appropriate cut-points applied to the time in intensity bands data, and 3) the minimum acceleration measured for a range of different durations, this is the MX metric where X refers to the duration. For example, $M60$ refers to the minimum acceleration for the most active 60 minutes of the day [53]. We also categorise all awake time into bout lengths for sedentary, light, and moderate to vigorous physical activity bouts using the categories defined by Gába et al [55]. After extracting sleep and physical activity, the remaining time within the 24-hour time cycle was classified as sedentary behaviour. Non-wear time and abnormally high acceleration values were detected using the data processing default setting [56]. These data were classified as missing and imputed using the accelerometer data from the same time interval on the remaining days of the week.

Covariates and moderators

The following individual participant data were obtained when available: age, sex, height, weight, waist circumference, maturational status, socioeconomic status (SES), ethnicity, screen time, sleep medications, and medically diagnosed conditions known to affect sleep. The country, city, time zone, daylight length, and season in which data were collected were also obtained.

Body mass index was converted to an age- and sex-standardised z-score. For participants under 18 years we used the CDC 2000 BMI-for-age growth reference, applied with the *childsds* package. No age-based growth reference applies from 18 years, so for adults we standardised body mass index against the adults in this sample. The two halves are consequently not the same quantity: below 18 the z-score is expressed relative to an external growth reference, and at 18 and above relative to this sample. We use the z-score in place of raw body mass index throughout, both as a covariate and as a moderator, so that a given value carries a comparable meaning at every age.

Statistical analysis

To reduce the risk of bias in secondary data analysis of IPD, we used a blind analysis approach [57]. We developed and tested models on a 10% holdout subsample of the dataset. When the analysis code was ready, we conducted the final analysis [58].

We screened all data for implausibly low or high measurements, as these likely indicate measurement device failures. Screening had three stages. First, we excluded values falling outside fixed physiological limits; for average acceleration this was 200 mg, a value above any sustainable 24-hour average. Second, because a miscalibrated accelerometer corrupts every variable derived from acceleration magnitude rather than isolated days, we excluded all physical activity variables for files whose mean average acceleration exceeded that limit (76 participants). Third, we excluded remaining values falling outside ±4 SD of the mean **of**

their own study, rather than of the pooled sample, so that the threshold reflected implausible measurement rather than between-study differences in device, protocol and population. The fixed limits were applied before the ±4 SD screen because the mean and standard deviation are themselves distorted by the values they are used to detect. Values excluded at any stage were treated as missing and imputed alongside genuinely missing data.

A day contributed to the analysis if it recorded more than 10 hours of wear time and, where sleep was detected, a sleep period longer than 200 minutes. Participants were additionally required to have at least 4 days of valid wear time. 345 participants, contributing 465 person-days, did not meet this criterion, leaving 24,264 participants and 148,749 person-days for analysis. Because the criterion is applied after imputation, the primary analysis and a sensitivity analysis that retains every eligible participant regardless of the number of valid wear days share the same imputations and the same standardisation constants, and are therefore directly comparable; the sensitivity analysis is reported in the supplementary materials.

We assumed missing covariate and moderator data to be missing at random or completely at random and imputed these missing values using multiple imputations by chained equations with the *mice* package in R [59] to provide 50 imputed datasets. Specifically, we used multilevel predictive mean matching imputation clustered by participants. We used Rubin's rule to pool the results from analyses conducted using each imputed dataset into one single set of results. We examined 95% confidence intervals for all analyses and used $P<0.05$ to assess for statistical significance. Both the intervals and the p-values are computed on the pooled Barnard-Rubin degrees of freedom, which account for the additional uncertainty introduced by imputation. We fitted 16 models for each moderator and report all of them without adjustment for multiple comparisons, so p-values close to 0.05 should be interpreted with corresponding caution. We used R (version 4.5.3) for all data cleaning and analysis. We report mean and 95% confidence intervals. For categorical

variables, percentages are reported. We examined model diagnostics from all models (see supplementary materials). Convergence and singularity are reported separately. A singular fit is one in which the optimiser succeeded but placed a variance component at zero, which indicates that the data do not support that random effect rather than that estimation failed.

To examine the relationship between sleep and physical activity (Research Questions 1-2) we ran mixed-effects models with random intercepts for study ID and for participant, accounting for the nesting of repeated days within participants and of participants within studies [60].

We predicted each sleep indicator by each physical activity indicator controlling for socioeconomic status, sex, and body mass index z-score. Quadratic terms were introduced in each model to allow for curvilinear relationships to be modelled. We interacted each predictor with age to reveal how the relationship between physical activity and sleep changes over the lifespan.

Models predicting physical activity from the preceding night's sleep (Research Question 3) use each sleep indicator lagged by one day, constructed within participants after eligibility screening so that the lagged value is always an eligible measurement from the immediately preceding calendar day. Person-days whose preceding day was absent from the data or did not meet the eligibility criteria therefore have no lagged value and do not contribute to these models, leaving 120,660 of 149,214 eligible person-days.

Protocol deviations

This paper addresses Hypotheses 1 and 2 from our registered protocol. We will address the remaining hypotheses in future research. To streamline our analysis and reduce redundancy, we made some changes to our protocol while working with the holdout data.

Using Study ID as fixed effect. We initially planned to use Study ID as a fixed effect, given that our study selection was not random. But, as we aimed for our findings to

generalise to other studies and populations, we instead opted to include random intercepts for Study ID which can better model generalisable patterns. This approach allows us to account for variability between studies without fully partialling out study effects, thereby better modelling common phenomena. While this change lessened the strength of model intercepts, it did not significantly alter model slopes. The results using Study ID as a fixed effect (i.e., those that align with the protocol) can be found in the supplementary materials and are qualitatively similar to the main results.

Sleep indicator variable. Our research examines various metrics of sleep: onset, duration, regularity, and efficiency. We initially planned to merge these variables into a single indicator to simplify analysis. However, no validated indicator existed which could be estimated from our measured variables. We decided against creating one as we could not validate it with our data. Consequently, our models illustrate the relationship between each aspect of sleep and physical activity, instead of describing a straightforward link between overall sleep and physical activity.

Log transformation. We aimed to improve model diagnostics using log transformations of skewed variables. Pilot analysis using holdout data revealed that while diagnostics improved on the log scale, model predictions followed extreme exponential patterns near the edge of the data range when back-transformed. Accordingly, we do not use log transformations in the main paper. However, these models can be found in supplementary materials.

Patient and public involvement

Patients and members of the public were not involved in the design, analysis or interpretation of this IPD study using secondary data. The results of this study have broad public health interest, particularly for policymakers. Findings from this study will be disseminated through institutional websites, press and media platform releases, and tailored messages to public health Organizations and governing bodies.

Results

The aggregated dataset describes 220,760 observations of daily physical activity and sleep from 28,823 unique participants. Of these, 71,546 days were excluded as they had insufficient wear time, which also resulted in the exclusion of 4,214 participants. A further 345 participants (465 days) were excluded for having fewer than 4 days of valid wear time, leaving 24,264 participants and 148,749 person-days. Table 1 shows demographic information for the included participants. Most participants came from Europe, Oceania, and South America, with limited data from other regions. There was a near-even split by sex (50.4% female), but age skewed towards both younger children (2-11 years, 44.9%) and older adults (66+ years, 25.1%).

Valid observations were not uniform across the days of the week ($\chi^2_{(6)} = 310.72$, p = < .001). Weekends were over-represented (Saturday: z = 6.08; Sunday: z = 11.62), while Wednesdays (z = -9.91) and Thursdays (z = -8.85) were under-represented. A table of study characteristics can be found in the supplementary materials.

The association between physical activity volume and sleep

We estimated the association between physical activity and sleep (RQ1) using mixed-effects models. These associations by age are presented in Table 2 and Figure 1. Higher physical activity volume was associated with longer sleep duration, higher sleep efficiency, earlier sleep onset, and more regular sleep. Although we observed statistically significant curvilinear relationships, these did not appear to meaningfully change the associations. While we observed statistically significant interactions between age and physical activity volume for sleep duration, efficiency, and onset, these were too small to be meaningful, and there was no evidence that the linear relationship with sleep regularity varied by age.

The association between physical activity intensity and sleep

We estimated how physical activity intensity is associated with sleep across different age groups and present the results controlling for sex, SES, and BMI, in Table 2 and Figure 2. We observed weak associations between physical activity intensity and sleep duration, sleep efficiency, and sleep onset, with a stronger relationship observed with sleep regularity. As with physical activity volume, we observed statistically significant curvilinear relationships for all sleep outcomes except sleep duration, but these were only meaningful for the association between physical activity intensity and sleep regularity, in which very low levels of intensity were strongly associated with poor sleep regularity and the benefits of increasing intensity diminished past one standard deviation above the mean. While age modified the relationship between physical activity intensity and sleep duration, efficiency, and regularity, the effect was negligible.

The association between of sleep duration and physical activity

We estimated the association between sleep duration and physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 3. At higher ages, both physical activity volume and intensity are lower than for lower ages. There was a weak curvilinear association between sleep duration and next-day physical activity volume and intensity, with physical activity highest following nights of slightly longer than average sleep; these associations weakened with age.

The association between sleep efficiency and physical activity

We estimated the association between sleep efficiency and physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 4. There was no linear association between sleep efficiency and physical activity volume, and only a weak positive association between sleep efficiency and more intense physical activity.

The association between sleep onset and physical activity

We estimated the association between sleep onset and physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 5. Earlier sleep onset was associated with higher next-day physical activity volume and, more weakly, intensity, although these associations were attenuated at older ages.

The association between sleep regularity and physical activity

We estimated the association between sleep regularity and physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 6. There was a positive linear relationship between sleep regularity and both physical activity volume and intensity. These relationships were slightly attenuated with age, such that for older adults the effect size was negligible.

Discussion

In this study, we collated and harmonised accelerometer-assessed physical activity and sleep data from 20 studies around the world, covering participants across the lifespan. We found that both the volume and intensity of physical activity were associated with longer sleep duration, higher sleep efficiency, earlier sleep onset, and more regular sleep. These relationships were the strongest for sleep regularity, although the relationship with physical activity intensity weakened with age. Although robust, these associations were small: changes of 1 SD in the physical activity metrics were associated with changes of 0.02 to 0.29 SD in the sleep metrics, with the largest associations observed for sleep regularity.

When examining the reverse direction, we observed that those who fell asleep earlier or at a more regular time engaged in more total and more intense physical activity the following day. Longer sleep duration conferred a smaller, curvilinear benefit for next-day physical activity, while sleep efficiency was only weakly associated with physical activity intensity and not with volume. Our sensitivity analyses suggest that these findings were

largely robust to the decisions made during analysis.

Our findings broadly align with meta-analyses of experimental research, although the strengths of the relationships we observed were much weaker [16,17]. For example, [16] found small ($0.22 - 0.30$) but significant relationships between physical activity and sleep duration and efficiency in adults using experimental studies for both acute and regular physical activity. It is difficult to directly compare these effect sizes, however, as the experimental studies directly manipulated physical activity, whereas our study examined the relationship between habitual physical activity and sleep. It may be that the routine nature of individuals' physical activity makes associations more difficult to detect. Trials also suggest that the benefits of exercise for sleep are concentrated in people whose sleep is poor at baseline [17,18], whereas our analysis did not target people with poor sleep. Similarly, most of the previous studies which have examined acute changes to physical activity were conducted in laboratory settings. These studies, while useful for understanding the potential mechanisms, are not ecologically valid and may not reflect the relationship between physical activity and sleep in the real world. Laboratory studies also tend to use more sensitive measures of sleep and physical activity, which may generate less statistical noise and make relationships easier to detect. It is important to note that sleep duration will sometimes be influenced by parameters external to the individual, such as work or school schedules or parenting duties, rather than by individual health behaviours.

Our findings can also be compared with observational studies of day-to-day sleep and physical activity. A meta-analysis of 33 such studies in adults did not support a bidirectional daily association: some sleep parameters were weakly associated with next-day physical activity, and more active individuals slept slightly less [24]. Two recent studies of consumer wearables, each with millions of person-nights, found that previous-day activity had little association with sleep [26], that shorter sleep was followed by more activity the next day [26,27], and that earlier-than-usual sleep onset was followed by more moderate-to-vigorous

activity [27]. Our finding that earlier sleep onset preceded more physical activity is consistent with these studies, but we found clearer associations between physical activity and that night's sleep, and physical activity was highest after slightly longer, rather than shorter, than average sleep. One explanation is that our models did not separate within-person (day-to-day) from between-person (habitual) associations, which can differ [61], so our estimates partly reflect stable differences between people. Neither wearable study examined sleep regularity, the outcome most strongly associated with physical activity in our data, and one suggested it as a direction for future research [27].

We found the most evidence for relationships between sleep regularity with physical activity volume and intensity. This relationship appeared to be bi-directional, with sleep onset and regularity influencing physical activity the following day, and physical activity influencing sleep onset and regularity that night. This is consistent with the idea that sleep and physical activity are mutually reinforcing behaviours [15], or that they are habitual and linked to factors such as the value placed on health. Consistent physical activity may also act as a non-photic zeitgeber that helps stabilise circadian rhythms [62]. Sleep regularity is a stronger predictor of mortality than sleep duration [63], and irregular sleepers remain at higher risk of cardiovascular events and depression even when they meet sleep duration recommendations [64,65], so the finding that physical activity is associated with sleep regularity may be particularly important. In pooled data from six cohorts, the combination of irregular sleep and low daily step counts was associated with higher adiposity and a less favourable lipid profile [66].

Contrary to expectations, we found little evidence to suggest that the associations of physical activity with next day sleep differed by age. While several of the analyses suggested statistically significant interactions between age and physical activity, these were too small to be clinically meaningful. This suggests that the interrelationship between physical activity and sleep are consistent across the lifespan, which is inconsistent with previous

meta-analyses conducted on different ages. A large study of adult wearable users similarly found that day-level associations were consistent across age groups [26]. More specifically, meta-analyses in adult populations have generally found associations [16] while those conducted in children and adolescents are less clear [15,21,22]. This would seem to suggest that the differences in findings across these meta-analyses are due to differences beyond just the age of the populations, such as the type of study designs used or the measures of physical activity and sleep. It also demonstrates the importance of using individual participant data meta-analyses to examine these types of questions.

These findings have implications for public health recommendations. Our results support promoting physical activity as a strategy for maintaining regular sleep patterns across the lifespan. This message fits with Australia's recent 24-hour movement guidelines for adults, which pair physical activity recommendations with consistent bed and wake times [67]. Given the well-established links between sleep and overall health [3,4], incorporating physical activity into public health messaging may have downstream benefits for population health. However, our findings suggest that physical activity is not a universal solution for improving all aspects of sleep. Public health messages should, therefore, emphasise the role of physical activity in promoting consistent sleep schedules rather than framing physical activity as a universal solution for improving sleep. Additionally, we did not examine these relationships specifically in individuals with sleep disorders or other health conditions, where the effects of physical activity may differ. Future research should explore whether targeting sleep regularity through physical activity interventions can provide meaningful health benefits beyond sleep itself.

Our study has several strengths, including the use of accelerometer-assessed measures, which reduce the bias associated with self-reported sleep and activity data, and the harmonisation of individual data from multiple studies, conducted in geographically diverse regions and among multiple age groups, allowing for robust and generalisable findings. We

also used modern methods for processing and estimating physical activity, including using the full 24 hours of data, and avoiding the use of intensity cut-points. However, the study is not without limitations. Despite our efforts to harmonise data, variability in accelerometer wear locations and slight differences in data collection protocols across studies may have introduced measurement inconsistencies; physical activity estimates from different wear locations are strongly correlated but agree less well in absolute terms, particularly between wrist and hip [68]. We also chose to maximise the data available for analysis, at the expense of some precision in the estimates. Specifically, we included participants with four or more days of valid wear time, while some recommend at least seven nights to estimate metrics such as sleep regularity [69], and reliable estimates of an individual's night-to-night variability in sleep may require more than a month of data [70]. However, the processing method we used specifically allows for comparisons to day-pairs and this methodology has been used by others [e.g., 63]. Sleep regularity scores also depend on the software used to calculate them, with GGIR producing scores around 20 points lower than the sleepreg package [71], so absolute values from our data should not be compared directly with studies that used other implementations. Additionally, although we controlled for several potential confounders, there may still be unmeasured factors affecting the observed associations. Finally, while our study captured data from across the lifespan, the limited representation of certain age groups, especially adolescents and young adults, restricts the generalisability of our conclusions to these populations. The limited research on those aged 18-35 is particularly concerning given that young adults are the most likely to report difficulty falling asleep [11].

Conclusion

Our study provides new insights into the relationship between physical activity and sleep across the lifespan using a large, globally diverse dataset with device-based measurements. We found that physical activity was most consistently associated with sleep regularity, with weaker and less consistent associations observed for sleep duration, efficiency,

and onset. Additionally, our analyses revealed beneficial associations, where higher physical activity levels were associated with greater sleep regularity, and more regular sleep patterns predicted higher physical activity the following day. However, this bidirectional association was attenuated in older adults, suggesting that the interplay between sleep and physical activity may change with age. Findings suggest that public health efforts to promote physical activity may be particularly valuable for improving sleep consistency rather than sleep duration or efficiency. While physical activity is unlikely to be a universal solution for sleep improvement, encouraging regular movement could support stable sleep patterns, which are increasingly recognized as a key determinant of overall health. Future research should explore how targeted interventions—particularly those addressing sleep regularity—may enhance both sleep and physical activity behaviours across different age groups and populations.

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Table 1

Participant characteristics

	Age group				
Characteristic	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Numeric variables					
N	10,857	1,921	308	5,040	6,066
Age	9.73 (1.49)	15.74 (1.72)	29.16 (5.07)	57.68 (7.37)	72.72 (4.52)
BMI	17.99 (3.19)	21.61 (3.76)	26.46 (6.42)	26.80 (4.79)	27.08 (4.31)
BMI z-score	0.25 (1.07)	0.09 (1.02)	-0.09 (1.38)	-0.02 (1.03)	0.05 (0.93)
Valid Weartime Hours	23.13 (1.60)	23.23 (1.54)	23.25 (1.61)	23.74 (0.82)	23.89 (0.40)
Valid Weatime Days	5.62 (1.89)	4.97 (1.34)	4.79 (1.31)	7.34 (2.48)	6.48 (2.09)
PA Intensity	-2.13 (0.19)	-2.23 (0.19)	-2.43 (0.19)	-2.50 (0.19)	-2.68 (0.22)
PA Volume	42.53 (24.27)	46.73 (14.30)	39.53 (10.69)	38.82 (11.60)	30.86 (9.71)
Sleep Duration (min)	486.90 (52.97)	416.72 (55.29)	421.10 (62.52)	408.42 (55.81)	404.16 (63.32)
Sleep Efficiency (%)	80.75 (7.40)	87.13 (6.30)	87.28 (6.12)	87.79 (5.52)	86.50 (6.03)
Sleep Onset (clock time; SD in h:mm)	20:55 (01:08)	23:30 (01:26)	23:40 (01:46)	23:41 (01:23)	00:23 (01:14)
Sleep Regularity	54.00 (12.73)	57.03 (12.15)	55.79 (12.76)	59.45 (11.35)	54.53 (12.49)
Accelerometer Wear Location					
Wrist	2,567 (23.64%)	1,883 (98.02%)	308 (100.00%)	5,040 (100.00%)	6,066 (100.00%)
Hip	8,290 (76.36%)	38 (1.98%)	-	-	-
Country					
Australia	1,505 (13.86%)	303 (15.77%)	108 (35.06%)	186 (3.69%)	-
Finland	508 (4.68%)	-	-	801 (15.89%)	-
New Zealand	455 (4.19%)	-	37 (12.01%)	151 (3.00%)	-
Spain	934 (8.60%)	5 (0.26%)	-	37 (0.73%)	2,386 (39.33%)
Switzerland	-	-	-	1,904 (37.78%)	1,023 (16.86%)

Table 1 continued

Characteristic	Age group				
	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
United Kingdom	447 (4.12%)	-	144 (46.75%)	1,961 (38.91%)	2,657 (43.80%)
Brazil	486 (4.48%)	727 (37.84%)	10 (3.25%)	-	-
Czech Republic	281 (2.59%)	848 (44.14%)	9 (2.92%)	-	-
Canada	522 (4.81%)	-	-	-	-
China	124 (1.14%)	-	-	-	-
Colombia	889 (8.19%)	-	-	-	-
Denmark	689 (6.35%)	-	-	-	-
Ecuador	981 (9.04%)	38 (1.98%)	-	-	-
India	574 (5.29%)	-	-	-	-
Kenya	525 (4.84%)	-	-	-	-
Portugal	702 (6.47%)	-	-	-	-
South Africa	506 (4.66%)	-	-	-	-
United States	729 (6.71%)	-	-	-	-
Ethnicity					
Non-White	2,127 (19.59%)	169 (8.80%)	14 (4.55%)	94 (1.87%)	240 (3.96%)
Unclear	4,433 (40.83%)	1,192 (62.05%)	152 (49.35%)	1,337 (26.53%)	2,424 (39.96%)
White	4,297 (39.58%)	560 (29.15%)	142 (46.10%)	3,609 (71.61%)	3,402 (56.08%)
Region					
Europe	3,561 (32.80%)	853 (44.40%)	153 (49.68%)	4,703 (93.31%)	6,066 (100.00%)
Oceania	1,960 (18.05%)	303 (15.77%)	145 (47.08%)	337 (6.69%)	-
South America	2,356 (21.70%)	765 (39.82%)	10 (3.25%)	-	-
Africa	1,031 (9.50%)	-	-	-	-
Asia	698 (6.43%)	-	-	-	-

Table 1 continued

	Age group				
Characteristic	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
North America	1,251 (11.52%)	-	-	-	-
Season					
Autumn	3,387 (31.20%)	446 (23.22%)	39 (12.66%)	1,241 (24.62%)	1,268 (20.90%)
Spring	2,687 (24.75%)	1,228 (63.93%)	124 (40.26%)	1,224 (24.29%)	1,989 (32.79%)
Summer	1,381 (12.72%)	85 (4.42%)	85 (27.60%)	1,270 (25.20%)	1,250 (20.61%)
Winter	3,402 (31.33%)	162 (8.43%)	60 (19.48%)	1,305 (25.89%)	1,559 (25.70%)
Sex					
Female	5,693 (52.45%)	1,002 (52.16%)	222 (72.08%)	2,714 (53.85%)	2,551 (42.05%)
Male	5,162 (47.55%)	919 (47.84%)	86 (27.92%)	2,326 (46.15%)	3,515 (57.95%)
Sleep Conditions Reported					
No	625 (5.76%)	-	-	3,511 (69.66%)	5,140 (84.73%)
Yes	50 (0.46%)	-	-	310 (6.15%)	626 (10.32%)
Not Reported	10,182 (93.78%)	1,921 (100.00%)	308 (100.00%)	1,219 (24.19%)	300 (4.95%)
Socioeconomic Status					
Low	2,306 (25.22%)	208 (11.56%)	2 (0.74%)	1,074 (21.98%)	2,388 (39.38%)
Medium	2,920 (31.94%)	884 (49.14%)	191 (70.74%)	2,407 (49.26%)	2,324 (38.32%)
High	3,916 (42.84%)	707 (39.30%)	77 (28.52%)	1,405 (28.76%)	1,352 (22.30%)

Note. For categorical variables the value is the count, and percentage. For numeric variables the value is the Mean and SD. Sleep onset is a clock time; its SD is a duration in hours:minutes. Total N = 24,264. 72 participants had missing age data and are counted in the total but not in any age band.

Table 2

Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

	Physical activity volume (z)				Physical activity intensity (z)			
Term	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
Sleep duration (z)								
(Intercept)	0.08 [-0.09, 0.25]	0.09	0.91	.364	0.08 [-0.08, 0.24]	0.08	1.00	.320
Physical activity	0.10 [0.09, 0.12]	0.01	11.23	< .001	0.09 [0.06, 0.11]	0.01	7.80	< .001
Age	0.00 [0.00, 0.00]	0.00	-3.32	< .001	0.00 [0.00, 0.00]	0.00	-3.45	< .001
Physical activity^2	-0.01 [-0.01, 0.00]	0.00	-3.43	< .001	0.01 [0.00, 0.02]	0.01	1.86	.063
Physical activity × Age	0.00 [0.00, 0.00]	0.00	-4.00	< .001	0.00 [0.00, 0.00]	0.00	-5.09	< .001
Age × Physical activity^2	0.00 [0.00, 0.00]	0.00	-0.85	.397	0.00 [0.00, 0.00]	0.00	-3.73	< .001
Sleep efficiency (z)								
(Intercept)	0.16 [0.00, 0.33]	0.08	1.98	.048	0.19 [0.01, 0.36]	0.09	2.11	.034
Physical activity	0.11 [0.10, 0.13]	0.01	14.52	< .001	0.07 [0.05, 0.09]	0.01	7.98	< .001
Age	0.00 [0.00, 0.00]	0.00	-0.56	.573	0.00 [0.00, 0.00]	0.00	-1.49	.136
Physical activity^2	-0.02 [-0.03, -0.02]	0.00	-12.10	< .001	-0.04 [-0.05, -0.03]	0.00	-7.59	< .001
Physical activity × Age	0.00 [0.00, 0.00]	0.00	-12.11	< .001	0.00 [0.00, 0.00]	0.00	-7.12	< .001
Age × Physical activity^2	0.00 [0.00, 0.00]	0.00	2.41	.016	0.00 [0.00, 0.00]	0.00	6.18	< .001
Sleep onset (z)								
(Intercept)	0.10 [-0.18, 0.37]	0.14	0.70	.485	0.10 [-0.17, 0.36]	0.13	0.72	.472
Physical activity	-0.02 [-0.04, -0.01]	0.01	-3.95	< .001	-0.06 [-0.07, -0.05]	0.01	-9.61	< .001
Age	0.00 [0.00, 0.00]	0.00	-1.65	.099	0.00 [0.00, 0.00]	0.00	-0.74	.457
Physical activity^2	-0.01 [-0.01, -0.01]	0.00	-6.43	< .001	-0.03 [-0.04, -0.02]	0.00	-8.22	< .001
Physical activity × Age	0.00 [0.00, 0.00]	0.00	-14.02	< .001	0.00 [0.00, 0.00]	0.00	1.84	.066
Age × Physical activity^2	0.00 [0.00, 0.00]	0.00	8.11	< .001	0.00 [0.00, 0.00]	0.00	10.79	< .001
Sleep regularity (z)								

Table 2 continued

	Physical activity volume (z)				Physical activity intensity (z)			
Term	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
(Intercept)	0.12 [-0.02, 0.27]	0.07	1.67	.096	0.28 [0.11, 0.45]	0.09	3.16	.002
Physical activity	0.25 [0.23, 0.26]	0.01	28.89	$< .001$	0.29 [0.27, 0.31]	0.01	28.95	$< .001$
Age	0.00 [0.00, 0.00]	0.00	0.45	.656	-0.01 [-0.01, 0.00]	0.00	-3.57	$< .001$
Physical activity^2	-0.02 [-0.03, -0.02]	0.00	-8.56	$< .001$	-0.11 [-0.12, -0.10]	0.01	-17.67	$< .001$
Physical activity $\times$ Age	0.00 [0.00, 0.00]	0.00	-1.33	.186	0.00 [0.00, 0.00]	0.00	-20.95	$< .001$
Age $\times$ Physical activity^2	0.00 [0.00, 0.00]	0.00	-8.31	$< .001$	0.00 [0.00, 0.00]	0.00	8.92	$< .001$

Note. Adjusted for SES, sex, and BMI z-score. Outcomes variables are listed in the column headers. Standardized (z-scores) are indicated with '(z)'.

Table 3

Sleep predicting physical activity controlling for SES, sex, and BMI z-score

	Physical activity volume (z)				Physical activity intensity (z)			
Term	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
Sleep duration (z)								
(Intercept)	1.17 [0.82, 1.52]	0.18	6.62	< .001	1.01 [0.88, 1.14]	0.07	15.09	< .001
Sleep duration	0.05 [0.04, 0.06]	0.00	11.10	< .001	0.04 [0.03, 0.04]	0.00	8.13	< .001
Age	-0.03 [-0.03, -0.02]	0.00	-38.57	< .001	-0.03 [-0.03, -0.03]	0.00	-47.70	< .001
Sleep duration^2	-0.03 [-0.04, -0.02]	0.00	-9.73	< .001	-0.02 [-0.02, -0.01]	0.00	-6.35	< .001
Sleep duration × Age	0.00 [0.00, 0.00]	0.00	-12.36	< .001	0.00 [0.00, 0.00]	0.00	-2.87	.004
Age × Sleep duration^2	0.00 [0.00, 0.00]	0.00	2.75	.006	0.00 [0.00, 0.00]	0.00	2.62	.009
Sleep efficiency (z)								
(Intercept)	1.16 [0.81, 1.50]	0.18	6.58	< .001	1.00 [0.87, 1.13]	0.07	15.01	< .001
Sleep efficiency	0.00 [-0.01, 0.01]	0.01	0.02	.981	0.01 [0.00, 0.02]	0.01	2.30	.022
Age	-0.03 [-0.03, -0.02]	0.00	-38.30	< .001	-0.03 [-0.03, -0.03]	0.00	-47.48	< .001
Sleep efficiency^2	-0.01 [-0.02, -0.01]	0.00	-6.26	< .001	-0.01 [-0.01, -0.01]	0.00	-4.59	< .001
Sleep efficiency × Age	0.00 [0.00, 0.00]	0.00	2.05	.041	0.00 [0.00, 0.00]	0.00	-1.52	.128
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	0.00	2.66	.008	0.00 [0.00, 0.00]	0.00	0.80	.423
Sleep onset (z)								
(Intercept)	1.18 [0.82, 1.54]	0.18	6.40	< .001	1.00 [0.86, 1.13]	0.07	14.48	< .001
Sleep onset	-0.14 [-0.15, -0.12]	0.01	-20.83	< .001	-0.07 [-0.08, -0.06]	0.01	-11.37	< .001
Age	-0.03 [-0.03, -0.02]	0.00	-39.21	< .001	-0.03 [-0.03, -0.03]	0.00	-47.94	< .001
Sleep onset^2	-0.03 [-0.04, -0.02]	0.00	-8.21	< .001	0.00 [-0.01, 0.00]	0.00	-0.84	.398
Sleep onset × Age	0.00 [0.00, 0.00]	0.00	13.43	< .001	0.00 [0.00, 0.00]	0.00	9.46	< .001
Age × Sleep onset^2	0.00 [0.00, 0.00]	0.00	4.31	< .001	0.00 [0.00, 0.00]	0.00	0.05	.963
Sleep regularity (z)								

Table 3 continued

	Physical activity volume (z)				Physical activity intensity (z)			
Term	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
(Intercept)	1.10 [0.77, 1.44]	0.17	6.50	$< .001$	0.97 [0.85, 1.09]	0.06	15.35	$< .001$
Sleep regularity	0.16 [0.15, 0.17]	0.00	38.12	$< .001$	0.12 [0.11, 0.13]	0.00	28.26	$< .001$
Age	-0.02 [-0.03, -0.02]	0.00	-37.25	$< .001$	-0.03 [-0.03, -0.03]	0.00	-46.89	$< .001$
Sleep regularity^2	0.01 [0.00, 0.01]	0.00	2.12	.034	0.00 [0.00, 0.01]	0.00	0.62	.536
Sleep regularity $\times$ Age	0.00 [0.00, 0.00]	0.00	-19.27	$< .001$	0.00 [0.00, 0.00]	0.00	-18.06	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	0.00	-1.23	.218	0.00 [0.00, 0.00]	0.00	-0.85	.396

Note. Adjusted for SES, sex, and BMI z-score. Outcomes variables are listed in the row headers.

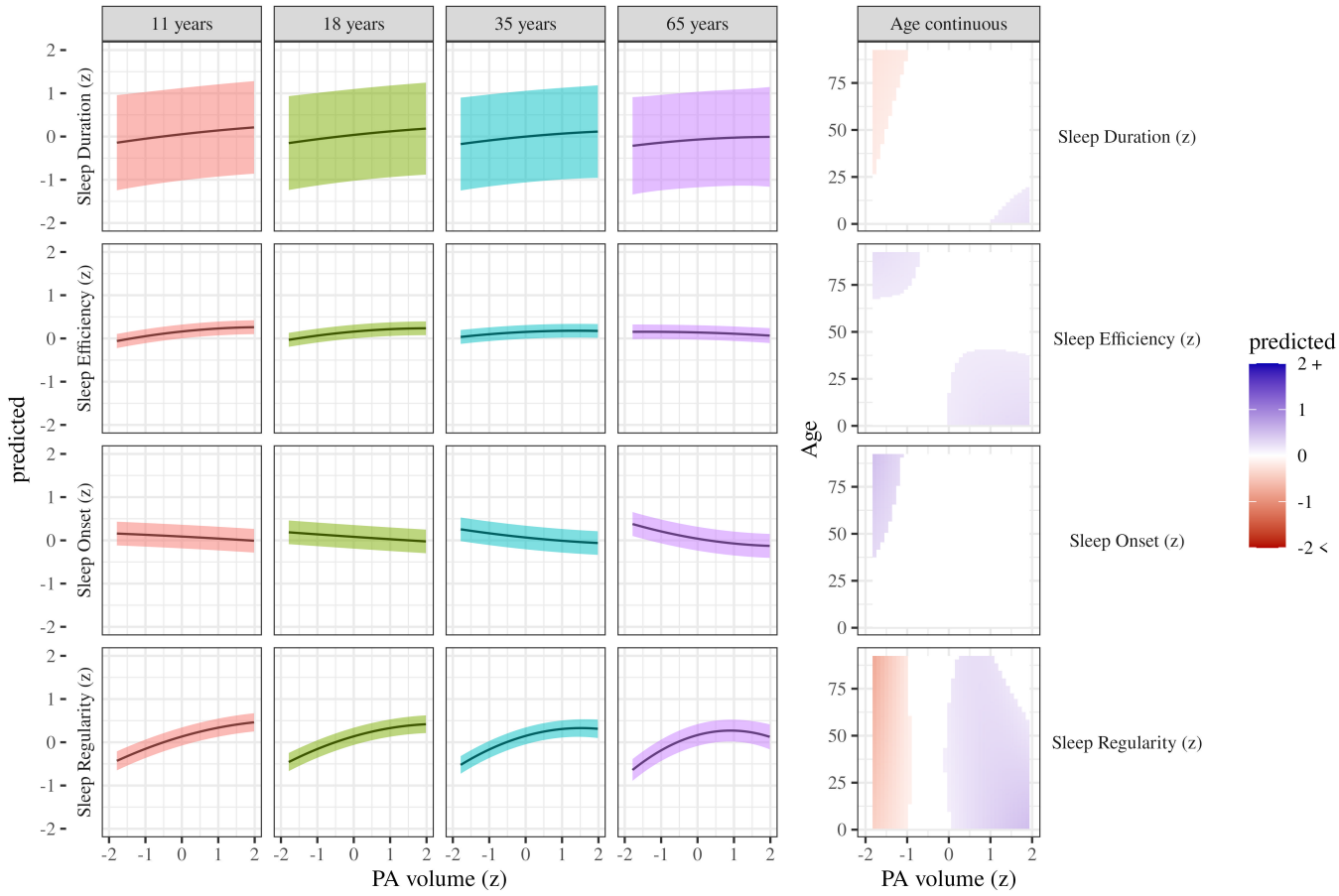


Figure 1. Standardized sleep metrics predicted by physical activity volume. The panels on the left show the curvilinear relationship between PA volume and each of the sleep outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero. These plots demonstrate exact turning points where the effects of PA volume change by age.

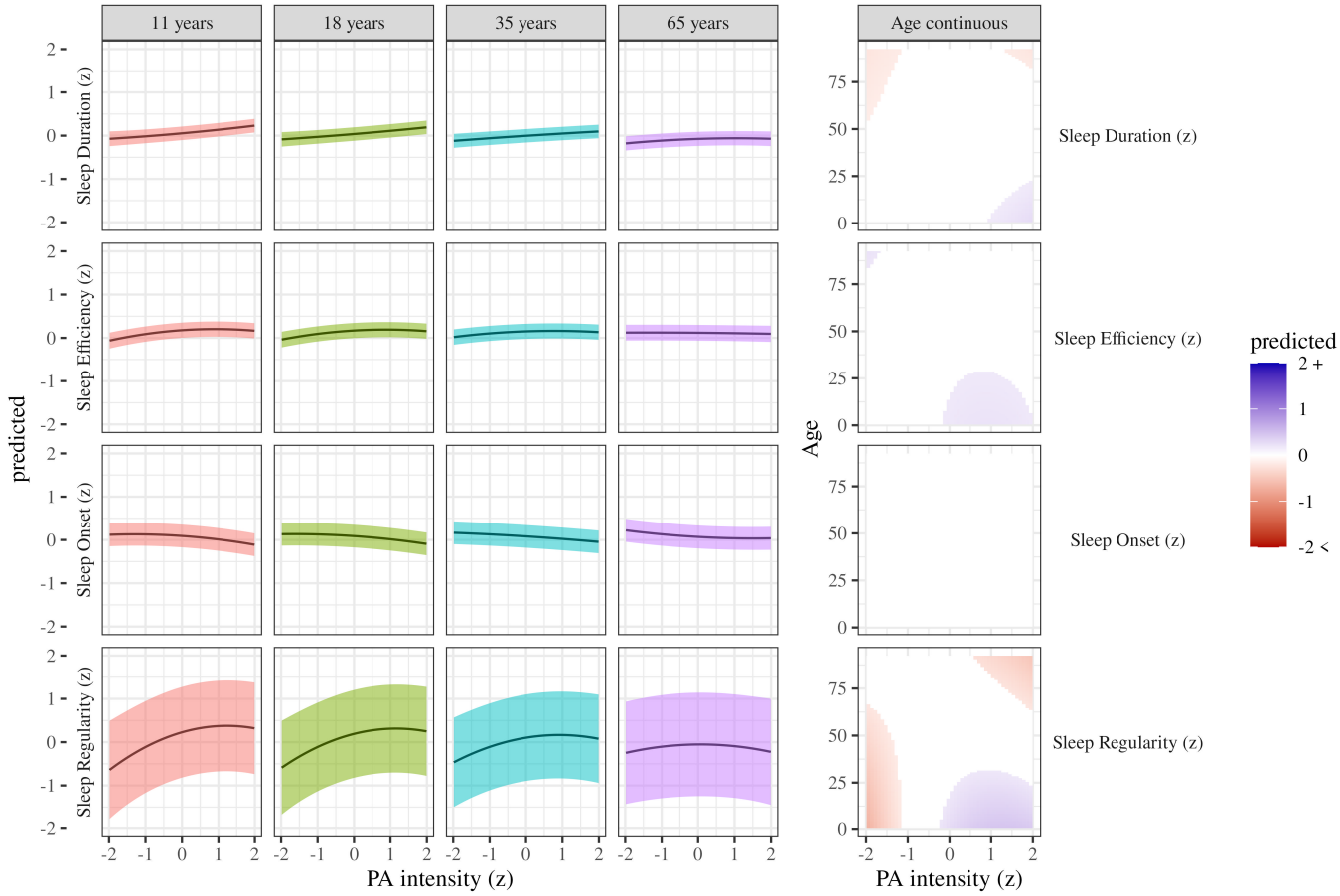


Figure 2. Sleep metrics predicted by physical activity intensity. The panels on the left show the curvilinear relationship between PA intensity and sleep outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

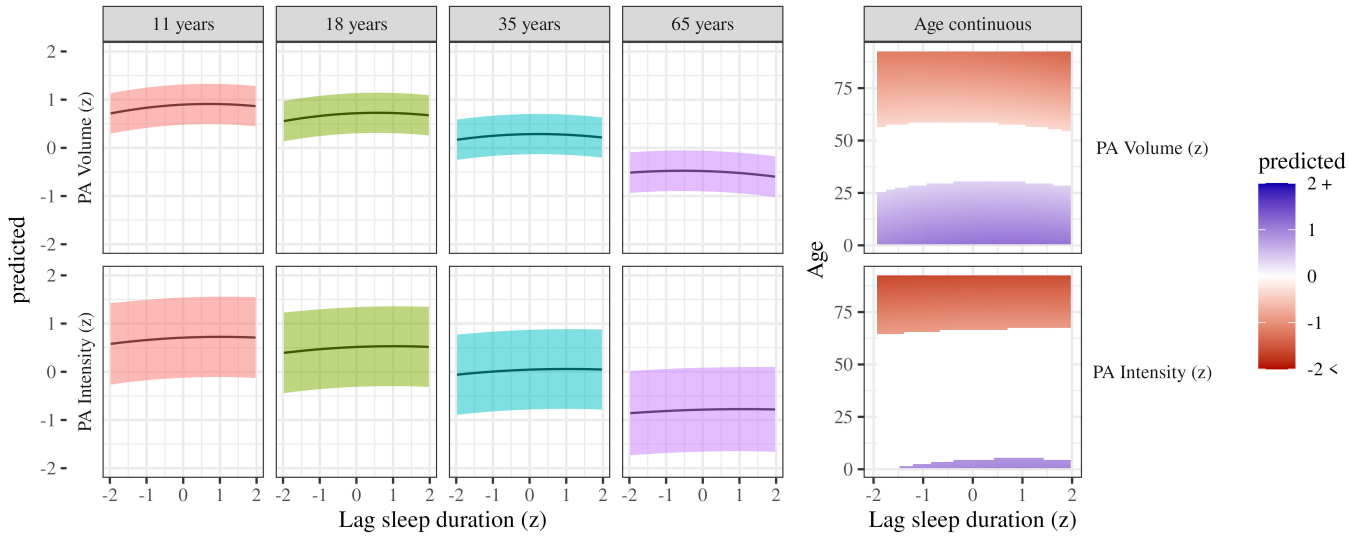


Figure 3. Physical activity is predicted by the duration of sleep the previous night. The panels on the left show the curvilinear relationship between sleep duration and the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

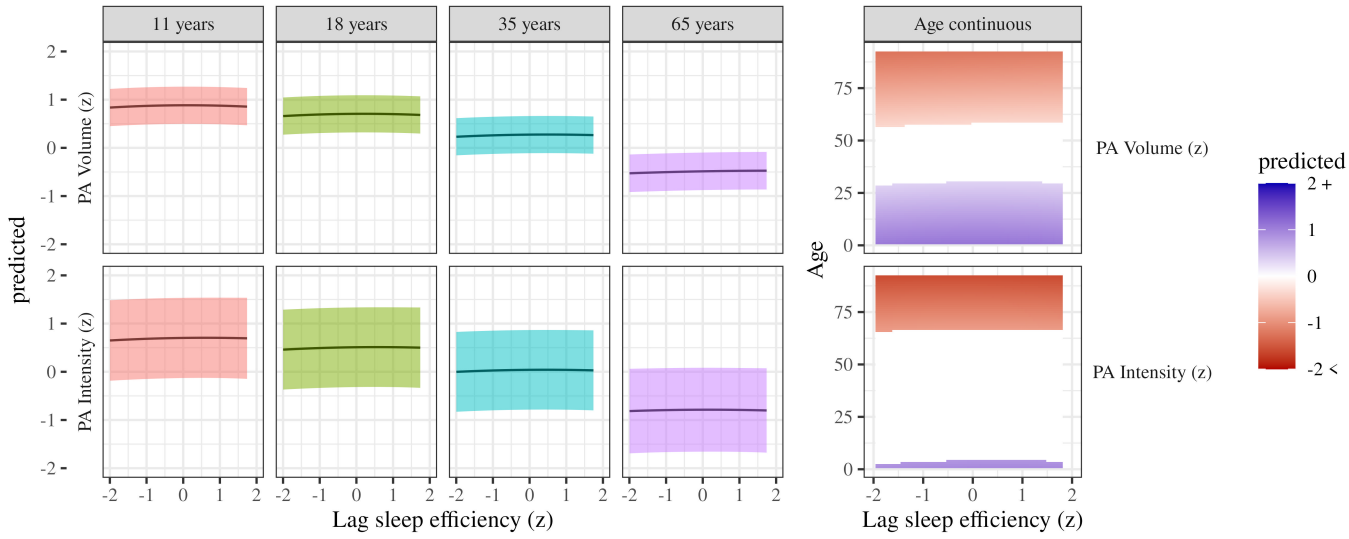


Figure 4. Physical activity predicted by previous night sleep efficiency. The panels on the left show the curvilinear relationship between sleep efficiency and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

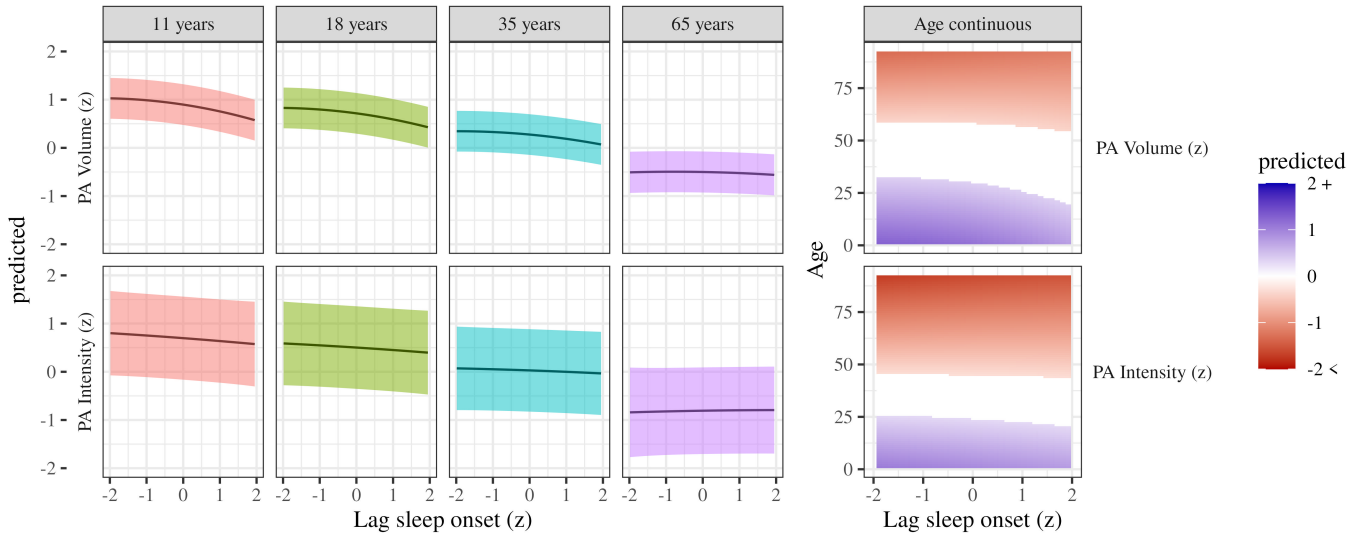


Figure 5. Physical activity predicted by previous night sleep onset. The panels on the left show the curvilinear relationship between sleep onset and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

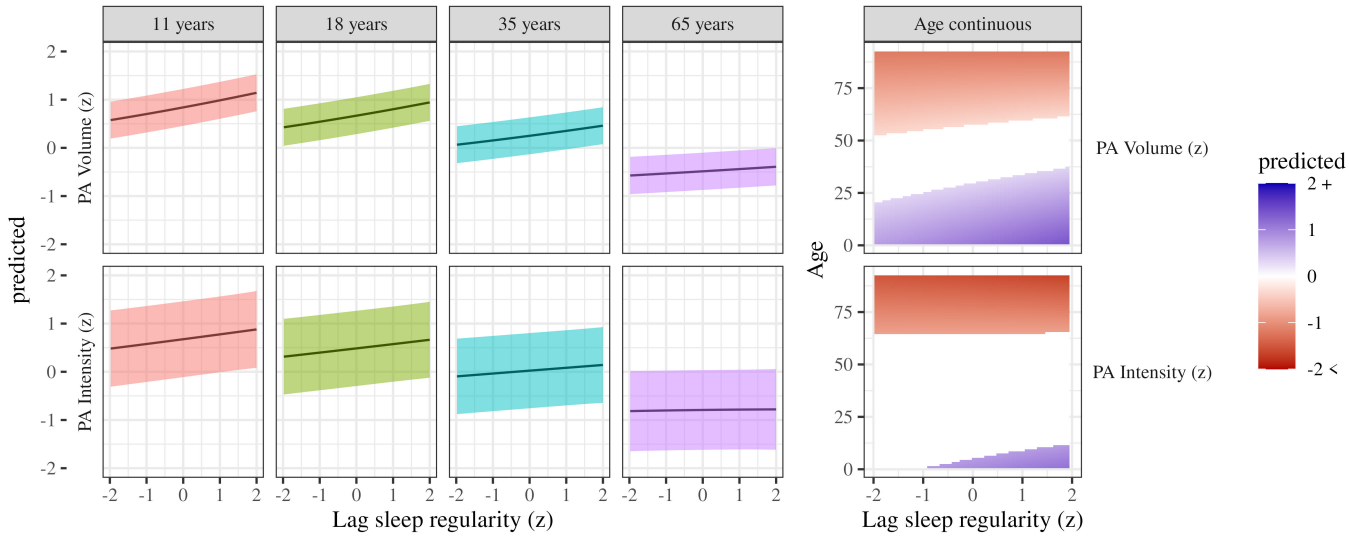


Figure 6. Physical activity predicted by previous night sleep regularity. The panels on the left show the curvilinear relationship between sleep regularity and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.